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RESEARCH ARTICLE

MSPE-DeiT: A Vision Transformer-Based Multimodal Framework for Skin Cancer Diagnosis

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ABSTRACT Skin cancer is one of the most common forms of cancer globally, with melanoma being the most fatal form. Early and precise detection is crucial for improving treatment outcomes, as timely disease management dramatically increases survival rates. This study presents a reliable multimodal deep learning framework that integrates dermoscopic images, 3D Total Body Photography scans, and patient clinical data to enhance diagnostic accuracy in skin cancer classification. The dataset comprised a combination of publicly available sources, including the recently released ISIC 2024 dataset. An XGBoost algorithm is utilized to get the predictions from the clinical data. For image-based analysis, this study proposes a Modality-Specific Patch Embedding Data efficient image Transformer (MSPE-DeiT) that uses a modality-specific patch embedding layer to train on the dermoscopic and 3D-TBP images. MSPE-DeiT is capable of capturing both local and global contextual information from high-resolution lesion images. Furthermore, this work uses an Artificial Neural Network (ANN) architecture for clinical data classification. Bayesian optimization is applied to fine-tune the ANN's hyperparameters, effectively mitigating vanishing gradient issues and improving model convergence. A Soft-Voting Ensemble classifier is used to integrate the results of all three models for better generalization on the unseen data. Experimental results demonstrate improvements in diagnostic accuracy on a multimodal held-out test set of around 80k images from the ISIC 2024 dataset, achieving an accuracy of 91.76%, an Area Under the Receiver Operating Characteristic Curve of 0.953, a specificity of 91.77%, and a sensitivity of 84.81%. The proposed approach is computationally inexpensive with an inference latency of 6.2 ms and a memory footprint of 83.59 MB, making it fit for deployment in clinical scenarios.

INDEX TERMS Skin cancer, ISIC, vision transformer, artificial neural network, XGBoost, Bayesian optimization, multimodal.

I. INTRODUCTION

Skin cancer is one of the most common forms of cancer globally, with melanoma being the most fatal form [1]. Early and precise detection is crucial for improving treatment outcomes, as timely disease management dramatically increases survival rates. Traditional diagnostic procedures, which rely heavily on clinical examination and dermoscopic evaluation by dermatologists, are fundamentally subjective [2]. As a result, there is growing interest in using automated computational methods to improve diagnostic accuracy and facilitate early diagnosis.

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With notable success in automated classification of skin lesions, deep learning, a branch of machine learning, has become a powerful instrument in medical image analysis [3]. By implicitly learning hierarchical feature representations from raw image data, Convolutional Neural Networks (CNNs) in particular have excelled in traditional image processing tasks [4]. CNN-based models have demonstrated the ability to perform dermatologist-level skin lesion classification in many studies, underscoring their potential to support clinical decision-making [5]. However, CNNs' intrinsic drawbacks, such as their limited receptive fields and dependence on local spatial features, can make it difficult for them to identify long-range relationships in complicated medical images [6].

To overcome these limitations, Vision Transformers (ViTs) have emerged as a possible alternative in medical imaging applications, such as skin cancer detection [6], lung cancer detection, brain disorders [7], etc. Unlike CNNs, which analyze images with localized convolutional kernels, ViTs use self-attention processes to record global contextual relationships throughout the image [8]. ViTs' capacity to model long-range relationships allows them to effectively evaluate complicated patterns in dermoscopic images, which is critical for discriminating between benign and malignant tumors [6]. Recent research has shown that ViTs can perform better than CNNs in medical image classification tasks, especially when trained on large datasets. Furthermore, ViTs improve interpretability because their attention maps reveal regions of interest that influence model predictions [6]. These benefits make ViTs a more realistic choice for automated skin cancer detection, furthering the field of deep learning-based dermatological analysis. Additionally, ensemble techniques are known to perform better than the individual constituents in a variety of machine learning tasks [9].

The International Skin Imaging Collaboration (ISIC) [10] has helped advance automated skin cancer diagnosis by supplying large-scale, annotated dermoscopic image collections. The yearly ISIC challenges have accelerated research in this field by defining benchmark tasks for lesion classification, segmentation, and detection. These competitions have boosted innovation by stimulating the creation of new architectures, ensemble learning algorithms, and ways for dealing with issues including data imbalance [11], [12], domain adaptability, and noise robustness [10].

Furthermore, by emphasizing practical limitations such as model generalizability across a range of populations and computing efficiency for deployment in resource-constrained environments, ISIC challenges have helped close the gap between research and clinical application. Consequently, deep learning models created through these contests have become stable, scalable, and extremely accurate for application in dermatological diagnosis.

Building on these advances, the main motivation behind this study is to propose a method for analyzing skin lesions using the proposed Modality-Specific Patch Embedding Data-efficient image Transformer (MSPE-DeiT), a Vision transformer-based architecture. The proposed methodology attempts to improve lesion characterization by utilizing self-attention mechanisms while keeping the computational practicality required for real-world application.

This study focuses on diagnosing skin cancer using a multimodal approach. The main contributions of the proposed methodology are as follows:

- Incorporated both dermoscopic and Three-Dimensional Total Body Photography (3D-TBP) scans
- Use of multiple datasets to improve generalizability
- Use of the ensemble approach to get the power of the image as well as the associated metadata
- Improved classification accuracy over the existing studies

The remainder of this study is divided as follows: Section II discusses the existing studies and the related work. Section III explains the proposed methodology for Skin Cancer classification, including the data collection and the various approaches used. The quantitative analysis and the various performance metrics used to evaluate the study, along with a critical analysis of the study, is presented in Section IV. Section V concludes the findings of the proposed work and discusses the limitations and future scope of the study.

II. RELATED WORK

Over the last decade, the field of automated skin cancer diagnosis has advanced dramatically, moving away from traditional image processing techniques and towards deep learning-based approaches, and further towards transformer-based approaches. This section discusses the latest deep-learning-based as well as transformer-based studies published in skin cancer detection and tries to identify the gaps in existing approaches.

A. DEEP LEARNING-BASED STUDIES

The advent of deep learning, particularly CNNs, revolutionized medical image analysis by enabling end-to-end feature learning directly from raw image data. Several studies have demonstrated the efficacy of CNNs in skin cancer classification, as discussed in this section.

Authors use variations of transfer learning models for the classification of the ISIC 2018 dataset, incorporating Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN) for image enhancement [13]. The variations of the proposed model achieved accuracies ranging from 83.20% to 85.80%.

The authors in [14] introduce InSiNet for detecting skin cancer that was tested using the ISIC 2018, 2019, and 2020 datasets. The approach outperformed other machine learning models, achieving accuracy as high as 94.59%. A Deep learning-based Skin Cancer Classification Network (DSCC_Net), is presented in [15] and is tested using the ISIC 2020, HAM10000, and DermIS datasets. With a 0.994 AUC and 94.17% accuracy, DSCC_Net surpasses baseline models, showing promise in skin cancer diagnosis.

Various experiments have been conducted on the HAM10000 dataset using CNNs. These include Gated Recurrent Unit (GRU) and Improved Orca Predation Algorithm (IOPA) [16], DenseNet169 and ResNet50 [17], and an attention-boosted Xception model [18], all demonstrating strong potential in medical image analysis and skin cancer diagnosis. A transfer learning model based on GoogLeNet [19] classified eight skin lesion types with 94.92% accuracy, further showcasing the effectiveness of advanced AI techniques in this domain. Similarly, a Deep Convolutional Neural Network (DCNN)-based approach using EfficientNet-B6 with transfer learning and metadata [20] achieved an AUC-ROC of 0.9681. Another hybrid model that combined machine learning with deep learning [21] outperformed dermatologists, reaching a classification accuracy of 93.00%.

B. TRANSFORMER-BASED STUDIES

Vision Transformers, which provide better global feature representation through self-attention techniques, have become more popular in medical imaging applications as a solution to the drawbacks of CNN-based designs [6].

A hybrid framework combining Multi-head Attention Block depthwise Separable Convolution Network (MABSCNET) and a modified Vision Transformer [22] with ensemble learning improves skin lesion detection. The approach was tested on multiple datasets, achieving 92.74% accuracy on ISIC 2020, 100.00% on ISIC 2018, and 94.24% on the Kaggle dataset, outperforming individual models.

SkinTrans, a vision transformer-based model [23], improves skin cancer classification with multi-scale feature extraction and self-attention, achieving 94.3% accuracy on HAM10000 and 94.1% on a clinical dataset. Another transformer-based [24] approach using image patch splitting and class rebalancing outperformed state-of-the-art models, achieving 94.1% accuracy on HAM10000 and excelling on the DERMOFIT dataset.

An improved Swin Transformer with a Hybrid Shifted Windows Multi-Head Self-Attention module (HSW-MSA) and a SwiGLU-based Multi-Layer Perceptron (MLP) [25] enhances skin cancer classification, achieving 89.36% accuracy on ISIC 2019. SkinViT [26], which integrates outlooker and transformer blocks, outperformed state-of-the-art models with 91.09% accuracy on ISIC 2019 and strong performance on other datasets.

An optimized vision transformer model with advanced preprocessing, segmentation, and feature extraction proposed in [27] achieved 99.81% accuracy on ISIC 2019.

The authors of [28] extracted features using Transformer encoder and ResNet, which were then fused to obtain the final classification results. They were able to achieve an Accuracy of 97.48% and Sensitivity of 87.53% on the ISIC 2019 dataset, while an accuracy of 96.86% and sensitivity of 96.12% on the PH2 dataset.

Gaps in existing approaches: Despite advances in automated skin lesion analysis, challenges remain. Vision Transformers have shown promise, matching or exceeding CNNs in dermatological image classification, particularly with large datasets, and offering enhanced interpretability through attention visualization. This study proposes a streamlined multimodal framework for skin cancer classification, aiming to optimize performance while maintaining architectural simplicity.

III. METHODOLOGY

This section outlines the methodology used in this study. Fig. 1 illustrates the complete workflow of the proposed approach.

A. DATA COLLECTION

The International Skin Imaging Collaboration (ISIC) has been collecting the dataset for skin cancer detection and classification from various clinical centers internationally, providing a dataset housing varied capture conditions [29]. For this study, the dataset is collected from various ISIC skin cancer datasets from 2016 to 2024. The datasets from ISIC 2016 to ISIC 2020 contain dermoscopic images, denoted as D_{id} while the ISIC 2024 dataset contains 3D-TBP scans, denoted as D_{it}, as well as the associated clinical data, denoted as D_c, for each lesion. Images from each of the datasets are denoted as $\{D_{ISIC16}, D_{ISIC17}, D_{ISIC18}, D_{ISIC19}, D_{ISIC20}, D_{ISIC24}\}$, for the datasets ISIC 2016, ISIC 2017, ISIC 2018, ISIC 2019, ISIC 2020, and ISIC 2024, respectively. The authors in [30] claim that the 3D-TBP image quality is comparable to the smartphones images, increasing the applicability of this dataset. In ISIC 2018, there are some ambiguous labeled data with respect to binary classification mapping, therefore such images are dropped from the analysis. Other labels are mapped to ''Benign'' and ''Malignant'' labels based on the label metadata as provided in the ISIC 2024 dataset. Table 1 lists the number of images used from each dataset.

TABLE 1. Number of images taken from various datasets.

Reference	Dataset	Malignant	Benign	Total
[31]	ISIC 2016	248	1,031	1,279
[29]	ISIC 2017	374	1,626	2,000
[32], [33]	ISIC 2018 [1]	1,927	9,415	11,342
[34]	ISIC 2019	6,989	24,159	31,148
[35]	ISIC 2020	584	32,542	33,126
[30]	ISIC 2024	393	4,00,666	4,01,059
Total		10,515	4,69,439	4,79,954

B. DATA PREPROCESSING

To ensure fairness in evaluation and prevent data leakage from the test set into the training process, the collected dataset is preprocessed using a stratified split with a 60:20:20 ratio to ensure that the class ratio and the modality ratio

[1] This dataset includes HAM10000 cases.

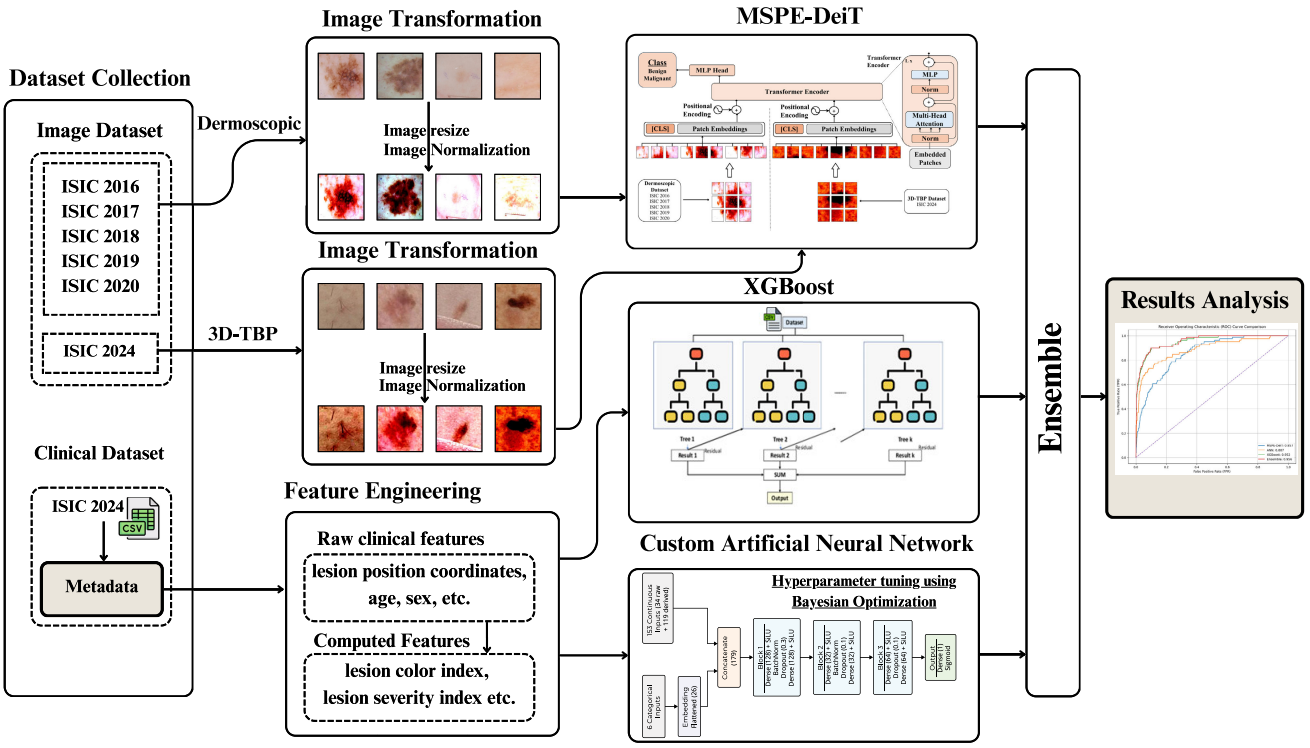


FIGURE 1. Proposed methodology.

(in case of image data) is consistent across the train-validation-test sets. The image data is resized to 224 × 224, the standard input size for Vision Transformers. The images are then preprocessed using normalization techniques [36] to produce the final input set, denoted as X_i. The corresponding ground-truth labels for D_i are denoted as L_{true}. For the clinical data D_c, missing values are imputed using the mode (most frequent value), and categorical columns are one-hot encoded to ensure compatibility with downstream machine learning classifiers.

C. FEATURE ENGINEERING

The raw clinical dataset D_c comprised 34 numerical features and 6 categorical features. Based on all these attributes, 119 new features were engineered to enhance the predictive performance of the downstream classifiers. These engineered features belong to various categories such as color-based metrics (e.g., lesion color index), composite indices (e.g., lesion severity index), and others, calculated from raw clinical attributes such as lesion position coordinates, age, sex, etc. [37]. These engineered features help capture relationships that could not be directly inferred from the raw data. Using mathematical transformations and combinations, additional features are derived, for example, the 3D position distance is calculated as $\sqrt{x^2 + y^2 + z^2}$. The final feature set used for model training is denoted as F_c, including both the raw features as well as the engineered ones. Incorporating these features significantly improved model performance, demonstrating the robustness and utility of this approach. This also helps reduce the complexity of the Artificial Neural Network (ANN) model, as some inter-relationships required for the final prediction were precomputed, simplifying the learned representation and facilitating faster convergence.

D. CLASSIFICATION OF CLINICAL DATA

This subsection discusses the models used for the classification of the clinical data of the subjects. As summarized in Algorithm 1, the clinical pipeline consists of two parallel classification branches: XGBoost and ANN.

Algorithm 1 Clinical Pipeline With XGBoost and ANN

Input: Clinical dataset D_c (from ISIC24)
Output: XgP, AnP (clinical predictions)
Feature extraction:
$F_c \leftarrow FeatureEngineered(D_c)$
XGBoost branch:
$XgP \leftarrow XGBoost(F_c)$
ANN branch with Bayesian Optimization:
$\theta^* \leftarrow \arg\min_{\theta \in \Theta} \mathcal{L}_{val}(\text{ANN}(F_c; \theta))$
$AnP \leftarrow \text{ANN}(F_c; \theta^*)$
return XgP, AnP

1) CLASSIFICATION USING XGBoost

For the classification of the clinical metadata D_c, eXtreme Gradient Boosting (XGBoost) is employed. XGBoost is a gradient-boosting algorithm that builds decision trees

sequentially, with each subsequent tree aiming to correct the errors of its predecessors [38]. The algorithm incorporates both L1 and L2 regularization to mitigate overfitting and improve model generalization. It also handles missing data natively, further enhancing model's efficiency. A class-balancing strategy is employed during model training to mitigate the extreme class imbalance in the dataset, ensuring stable optimization and meaningful discrimination of the minority class.

The final prediction from the XGBoost branch is denoted as $XgP = XGBoost(F_c)$, where F_c are the engineered features. Performance is evaluated using the Accuracy, Specificity, Sensitivity and the Area Under the Receiver Operating Characteristic (AUROC) Curve, with particular emphasis on sensitivity to malignant cases, ensuring the model's robustness in identifying critical instances.

2) CLASSIFICATION USING ANN

The categorical features were ordinally encoded, followed by a learnable embedding layer to ensure that the ANN learns the optimal representation for each category. To obtain the optimal classification results on the clinical metadata F_c, the ANN branch is trained with hyperparameter tuning using Bayesian Optimization. Formally, the optimal set of parameters is obtained as $\theta^* \leftarrow \arg\min_{\theta\in\Theta} \mathcal{L}_{val}(\text{ANN}(F_c;\theta))$. The tuned hyperparameters include the number of blocks, the skip connections, the dropout rate, and the optimizer used during training. Skip connections were incorporated in this design to check if the identity mappings preserved by the skip layers could help the ANN to learn the data better. Although, the results showed that skip connections did not help in improving the performance of the ANN model, and were eventually not retained in the final model as per the configuration selected using the Bayesian Optimization process.

The optimal parameters obtained using the Bayesian Optimization with a trial budget of 100, were used to produce the final prediction scores by the ANN model, denoted as *AnP*. Table 2 details the hyperparameters and the corresponding search space for the Bayesian Optimization process.

The ANN model was trained for 100 epochs with early stopping (patience = 12) and adaptive learning rate scheduling (reduction factor = 0.5, patience = 6). Adam optimizer was used for training the ANN model, due to its inherent and known ability to converge quickly and its proven performance in a variety of tasks. The final structure of the ANN consisted of three blocks, each block including two dense layers with the same number of nodes in each, including optional Batch Normalization and Dropout layers. Each block had an optional skip layer, connecting the output of the block to its input. To mitigate the class imbalance present in the dataset, focal loss [39] is used during the training. The detailed architecture diagram of the tuned model is illustrated in Fig. 2.

TABLE 2. Hyperparameter tuning search space for Bayesian optimization.

Hyperparameter	Search Space
Number of blocks	1-4
Activation function of each layer	ReLU, SiLU
Dropout	0.1-0.4
Skip Connection	True/False
Number of Nodes in each block	32-128
Batch-normalization	True/False
Focal Loss Alpha	0.75-0.99
Focal Loss Gamma	1-3
Learning Rate	10^{-5}-10^{-2}

ReLU - Rectified Linear Unit, SiLU - Sigmoid Linear Unit

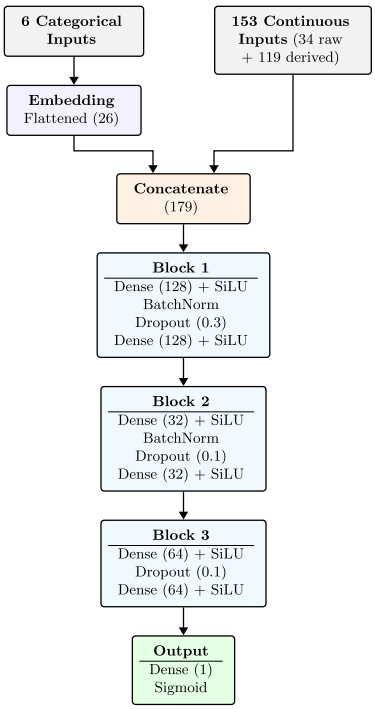


FIGURE 2. Architecture diagram of the proposed ANN model.

E. IMAGE CLASSIFICATION USING MSPE-DeiT

To efficiently handle the multimodal image dataset, MSPE-DeiT is designed to make use of a modality-specific patch-embedding and positional encoding layer, followed by a shared encoder and a classifier head. In this way, the model can learn from both the dermoscopic as well as 3D-TBP images, giving it an added advantage. To test the efficacy of the approach, three different base models are used for comparison, namely ViT-tiny, ViT-small [40], and DeiT-tiny [41]. The larger variants of these models need much larger datasets to train, hence they are excluded from the analysis. To speed up the convergence of the trained models, the models are initialised with pre-trained weights from the ImageNet dataset [42], which are subsequently fine-tuned on the collected dataset to obtain optimal results. The detailed architecture diagram of MSPE-DeiT is illustrated

Algorithm 2 Image Pipeline Using MSPE-DeiT

Input: Image datasets
$\{D_{ISIC16}, D_{ISIC17}, D_{ISIC18}, D_{ISIC19}, D_{ISIC20}, D_{ISIC24}\}$
Output: *ImP* (image-based prediction)
Merge and balance:
$D_{id} \leftarrow$
{dermoscopic images from $D_{ISIC16}, \ldots, D_{ISIC20}$}
$D_{it} \leftarrow$ {3D-TBP images from D_{ISIC24}}
$D_i \leftarrow D_{id} \cup D_{it}$
$X_i \leftarrow \text{ResizeNormalize}(D_i)$
$L_{true} \leftarrow$ Ground-truth labels corresponding to D_i
foreach *image* $x \in X_i$ **do**
  if $x \in D_{id}$ **then**
    $x_{\text{cls}} \leftarrow \text{Class Token}_{\text{dermo}}$
    $z_p \leftarrow \text{Patchify}_{\text{dermo}}(x)$
    $z_e \leftarrow \text{Embed}_{\text{dermo}}(z_p)$
  else
    $x_{\text{cls}} \leftarrow \text{Class Token}_{\text{3D-TBP}}$
    $z_p \leftarrow \text{Patchify}_{\text{3D-TBP}}(x)$
    $z_e \leftarrow \text{Embed}_{\text{3D-TBP}}(z_p)$
  $z_0 \leftarrow [x_{\text{cls}}; z_e] + Embed_{pos}$
  // At each layer l,
    $z_l = [z_l^{cls}, z_l^1, \ldots, z_l^N]$
  $L \leftarrow$ number of Transformer encoder layers
  for $l = 1$ **to** L **do**
    $z'_l \leftarrow (\text{LayerNorm}(z_{l-1}) +$
     $\text{MultiHeadSelfAttention}(\text{LayerNorm}(z_{l-1})))$
    $z_l \leftarrow z'_l + \text{FeedForwardMLP}(\text{LayerNorm}(z'_l))$
$ImP \leftarrow \text{Softmax}(FeedForwardMLP(z_L))$
return *ImP*

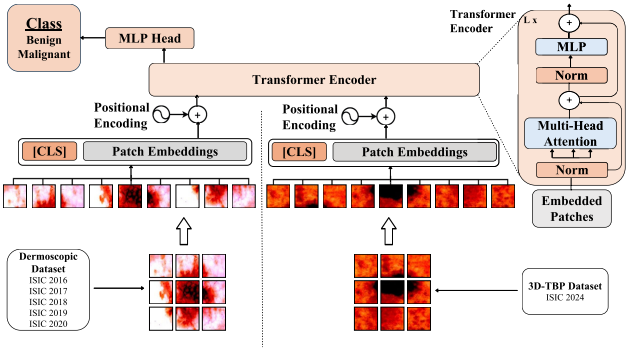


FIGURE 3. Architecture diagram of MSPE-DeiT.

in Fig. 3. The detailed process flow of the images is given in Algorithm 2. Class-balanced focal loss [43] is used during the training process to mitigate the inherent class imbalance present in the dataset.

Out of the 3 variants chosen for the analysis, the DeiT-Tiny architecture, with a patch size of 16×16, is selected, as it has demonstrated remarkable performance on smaller datasets, such as the one used in this study. DeiT's exceptional feature extraction capabilities and ability to learn from limited data make it particularly well-suited for tasks like medical image classification [44].

F. ENSEMBLE MODEL

A Soft-Voting Ensemble Classifier is used to combine the results of the 3 individual models trained on the varied datasets and training regimes. Isotonic regression was applied on the validation set to ensure that the probabilities of the three models are adequately calibrated. To get the optimal weights for combining the predictions, Sequential Least Squares Programming (SLSQP) [45] is employed that uses linear programming to find the optimal set of weights. The weights for the models MSPE-DeiT, XGBoost, and ANN are denoted as w_{MSPE}, w_{XGB}, and w_{ANN}, respectively. The ensemble model has the capability of combining the strengths of multiple learners to get an even better and strong model, thus avoiding overfitting and improving generalization of the model. This approach helps to predict correctly even if one of the models misclassified the given input, thus reducing the overall probability of misclassification and improving the model performance. The ensemble strategy combining the outputs of Algorithms 1 and 2 is illustrated in Algorithm 3.

Algorithm 3 Final Ensemble Decision

Input: *ImP*, *XgP*, *AnP*
Output: Final prediction $Y \in [0, 1]$, Probability of Malignant class

$$Y \leftarrow w_{\text{MSPE}}\,\text{ImP}^{(1)} + w_{\text{XGB}}\,\text{XgP}^{(1)} + w_{\text{ANN}}\,\text{AnP}^{(1)},$$

$$\text{subject to } w_{\text{MSPE}} + w_{\text{XGB}} + w_{\text{ANN}} = 1, \qquad w_i \geq 0.$$

return *Y*

G. EVALUATION METRICS

For a fair comparison of the performance results on the ISIC dataset, the dataset authors suggest the use of the AUROC score, which aims to check how well the model is able to separate the two classes in terms of its predicted probability, in contrast to other metrics that use fixed thresholds to compare the model performance. This study evaluates the proposed approach on the basis of AUROC as well as threshold-based metrics such as Accuracy, Specificity, and Sensitivity.

Accuracy measures the number of correct predictions out of the total predictions as given in (1) where TP, FP, TN, and FN represent the number of True Positives, False Positives, True Negatives, and False Negatives, respectively. Sensitivity measures the model's ability to correctly classify an instance that actually belongs to a positive class as given in (2). Specificity measures the model's ability to correctly classify an instance that actually belongs to a negative class as given

in (3).

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{2}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{3}$$

IV. RESULTS AND DISCUSSION

The data was split into 3 sets i.e., training set (60% of the total data), validation set (20% of the total data) and test set (20% of the total data). All the trained models in each branch were evaluated on the validation set for hyperparameter tuning and model selection. The final selected models were then evaluated on another held-out test set to avoid any optimism bias or inflated metrics. This section details the results obtained on the held-out test set for all the trained models. The held-out test set contains 80,212 records for the clinical data, and 95,990 records for MSPE-DeiT. The increased test set for MSPE-DeiT is due to the merging of the other dermoscopic datasets contributing 15,778 records into the test set. All these experiments were carried out on a single machine with the following specifications: Intel i9 32-Core Processor, 64 GB RAM and an NVIDIA RTX ADA 4000 GPU. A random seed value of 42 was used across all the experiments to ensure the reproducibility of the experiments carried out.

A. MSPE-DeiT

The modality-specific patch embedding (MSPE) layer was validated using an ablation study in which we compared the performance of three different architectural variants. These are (i) *Shared Embedding Layer* - where all input modalities have a shared latent space that projects them into one common latent space; (ii) *Separate Models for Each Modality* - where we trained separate, independent DeiT-based models per modality and aggregated their predictions; (iii) Our Proposed *MSPE Layer*, which contains separate, dedicated, trainable projection layers for each input modality prior to the Transformer Encoder. Quantitative results are presented in Table 3.

TABLE 3. Ablation study for the modality-specific projection embedding layers.

Architecture	AUROC	Accuracy@95 (%)	Sensitivity@95 (%)
Shared Embedding Layer	0.954	94.54	74.13
Separate models (Ensemble)	0.915	94.40	67.09
MSPE (Ours)	**0.965**	**96.54**	**75.94**

Accuracy@95 and Sensitivity@95 are the accuracy and sensitivity at specificity of 95%.

Our model with the proposed MSPE architecture achieved the highest AUROC of 0.965 and the highest Sensitivity@95 of 75.94%, indicating that our model outperformed other models in terms of its ability to discriminate between the malignant and the benign samples. In clinical settings, missing a false negative (i.e., a malignant sample being classified as benign) carries a higher risk than a false positive (i.e., a benign sample being classified as malignant), so it is imperative to catch every malignancy.

The Shared Embedding Layer variant achieved an accuracy@95 of 94.54% and Sensitivity@95 of 74.13%, which are close to the proposed model, suggesting that the shared layer also projected the input features in a smooth manner. To compare the performance of the three model architectures more rigorously, we performed statistical significance testing on the AUROC values using the DeLong test [46]. The DeLong test is a non-parametric comparison for correlated ROC curves that can be used to determine if the performance of our MSPE model is statistically significantly better than the other two model architectures. The results of this study are presented in Table 4.

TABLE 4. DeLong test for pairwise AUROC comparison against the proposed MSPE model.

Metric	MSPE (Ours)	Shared Embedding	Ensemble
AUROC	0.965	0.954	0.915
Z-value	–	4.222	16.433
***p*-value**	–	2.42×10^{-5}	< 0.0001

The DeLong test confirms that the AUROC of the proposed MSPE model is statistically significantly higher than both the Shared Embedding Layer ($Z = 4.222$, $p = 2.42 \times 10^{-5}$) and the Ensemble baseline ($Z = 16.433$, $p < 0.0001$), with both comparisons reaching the $p < 0.001$ significance threshold. These results validate that the marginal numerical gap in AUROC between MSPE and the Shared Embedding variant is not attributable to random variation.

All the metrics clearly indicate that the proposed MSPE variant is better than the other counterparts for this task. The standardized metrics, i.e. Accuracy@95 and Sensitivity@95, help in providing a threshold-invariant and fair comparison of the three variants, along with the statistical significance established by the DeLong test.

Fig. 4 compares different variants of MSPEs on the test data. We selected three variants for analysis: ViT Small (22M parameters), ViT Tiny (5.6M parameters), and the DeiT (5.7M parameters) [41]. These models were chosen for their lightweight nature and ability to achieve remarkable results with limited data. As evident from the figure, DeiT Tiny showed optimal performance, with a slightly higher AUROC value (0.965), as compared to ViT Tiny (0.964) and ViT Small (0.957). Though these differences are well within the margin of random error, it can be inferred that DeiT Tiny is a better choice due to its lightweight nature and ability to achieve better results with limited data. The first section of Table 5 shows how the different variants of MSPE performed on the given test set.

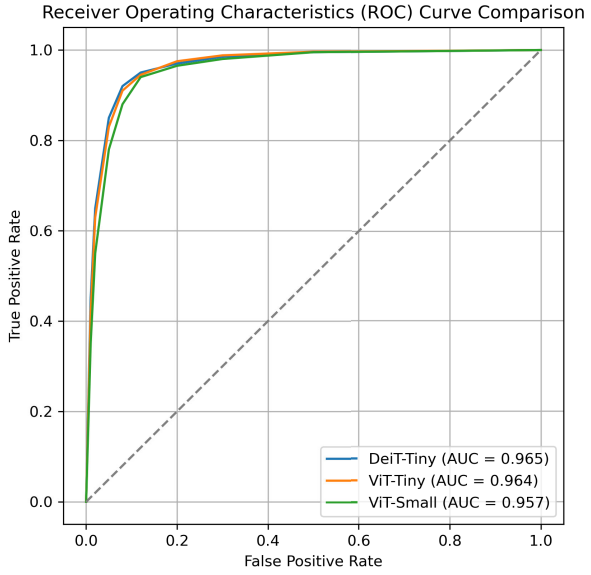


FIGURE 4. ROC curves for the various MSPE variants: DeiT-Tiny (proposed), ViT-Tiny, and ViT-Small.

TABLE 5. Comparison of architecture classes on different input modalities.

Architecture Class	Model	Input Modality	AUROC Score
MSPE Variants	deit-tiny-patch16-224	Dermoscopic Image + 3D-TBP	**0.965**
	vit-tiny-patch16-224	Dermoscopic Image + 3D-TBP	0.964
	vit-small-patch16-224	Dermoscopic Image + 3D-TBP	0.957
ANN Variants	KAN	Clinical	0.575
	ANN	Clinical	**0.930**
ML Models	XGBoost	Clinical	**0.952**
	Hist-Gradient Boosting	Clinical	0.934
	LightGBM	Clinical	0.923
	Extra Trees	Clinical	0.926

The proposed architecture is able to obtain a dual advantage using such dual embedding layers and a shared encoder architecture. The dermoscopic datasets have a fair class balance but suffer from a smaller number of images, making it difficult for Transformer-based models to capture complex relationships present in the data. On the other hand, the ISIC 2024 dataset, comprising 3D-TBP scans, has a huge number of images, but brings in an extreme class imbalance of 1000:1, making it difficult for the model to learn the patterns present in the minority class. Subjecting both modalities to a shared encoder and combined training regime helps the model overcome the problems posed by both the modalities, making it a better fit over training over individual modalities. Also, ViTs were chosen for this task instead of CNNs since they can process the entire image in parallel and are capable of capturing global contextual information through a self-attention mechanism. This is particularly useful in skin lesion images, where global information such as the lesion characteristics, border irregularity, and relationship between different parts of the images (captured through the self-attention mechanism of ViT) play a vital role in determining the class of the image. When comparing the three variants of ViTs for this task, DeiT-tiny was able to achieve relatively higher AUROC scores due to a number of advantages it possesses over the other models in the context of our dataset. Moreover, given the limited size of the malignant class with only around 10k images (only 393 for 3D-TBP), models like ViT-tiny and ViT-small seemed to slightly underperform due to their lack of capabilities to learn complex relationships from lesser number of instances.

Fig. 5 illustrates the training and validation loss curves for MSPE-DeiT. For the first 5 epochs, the encoder for the model was frozen, and only the embedding layers and the classifier head were trained. During this phase, the model quickly learned to distinguish between the two modalities, as evident by the initial dip in the validation loss. However, as the training progressed and the encoder was unfrozen, the model's performance further improved, with both the training and validation losses reducing from around 0.4 to around 0.34. This suggests that the encoder was able to learn relevant features from the combined data, i.e., the skin lesions from dermoscopic and 3D-TBP images. Moreover, the difference between the training and validation losses remained small throughout the training process, indicating that the model was not overfitting to the training data. The small gap (ranging from 0.05 to 0.15) between the training and validation losses also suggests that the model generalized well to unseen data.

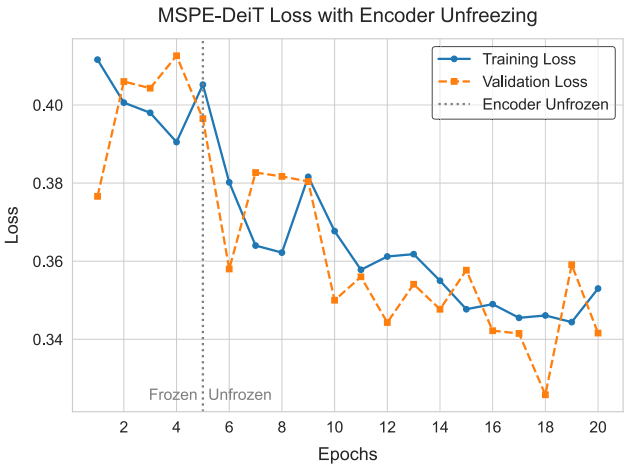


FIGURE 5. Training and validation loss curves for MSPE-DeiT on the image dataset.

B. ANN

Fig. 6 compares different neural network architectures using metadata. Two models were considered: ANNs and Kolmogorov-Arnold Networks (KANs) [47]. In the case of ANNs, the figure shows ROC curves for the 5 best models selected based on the AUROC scores on the validation set. KANs have recently gained attention due to their learnable activation functions, which aim to make them more powerful than traditional ANNs. However, as shown in the Fig. 6, traditional ANNs, when subjected to extensive hyperparameter tuning, can outperform KANs. Following hyperparameter tuning, the best selected ANN achieved an AUROC of 0.930 on the test set, a substantial improvement over the KANs (0.575). The second section of Table 5 summarizes the performance of the ANNs and KAN models on the test set.

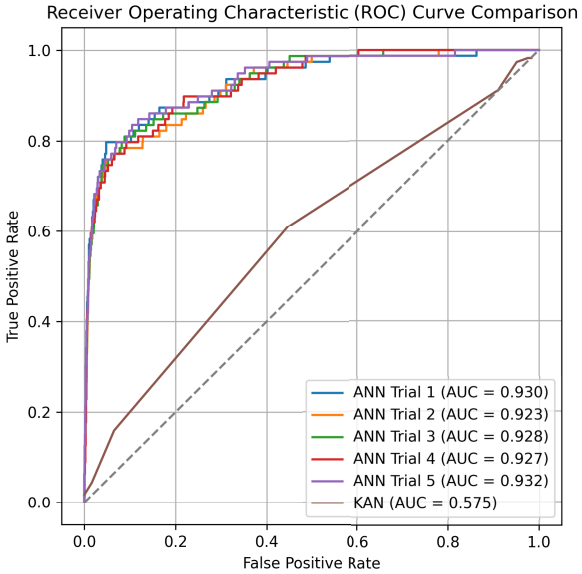


FIGURE 6. Test ROC curves for the ANN and KAN models, where the top 5 ANN trials are selected using the AUROC scores on the validation data.

Bayesian Optimized ANNs were able to outperform KANs due to several key factors. KANs dynamically learn the activation functions from the dataset instead of using pre-defined ones, making them highly data-hungry. In this study, the severe class imbalance (1000:1) provided insufficient representative data for the malignant class. Consequently, KANs struggled to correctly learn the activation functions for the minority class, leading to severe overfitting and poor generalization. In contrast, ANNs possessing a large hyperparameter space (number of layers, the number of nodes, and the activation function in each layer) significantly benefited from Bayesian optimization. This automated technique helped in determining the most optimal set of hyperparameters for the model, which might be infeasible through manual tuning or random search.

The training and validation loss curve for the final ANN model is given in Fig. 7. As evident from the figure, the model converged well with minimal gap between training and validation loss, indicating that the model generalizes well to the unseen data.

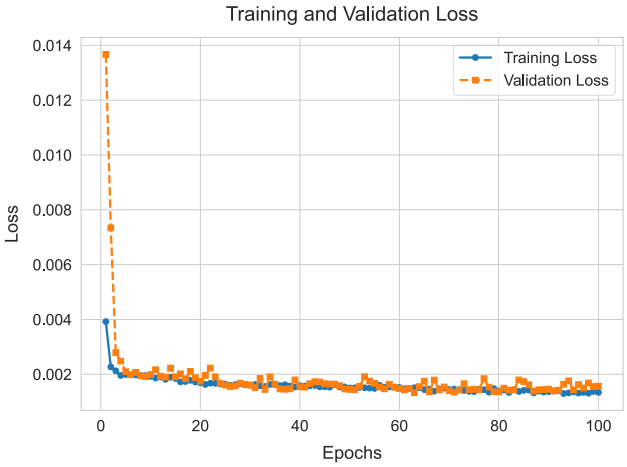


FIGURE 7. Training and validation loss curves of ANN on the clinical dataset.

C. XGBoost

Fig. 8 compares the performance of various Machine Learning (ML) models on the metadata. All these models were subjected to hyperparameter tuning using Grid Search with 5-fold cross validation to get the optimal set of hyperparameters. It can be observed from the figure that the AUROC score is the highest for the XGBoost model (0.952), as compared to the other trained models, namely, LightGBM (0.923), Extra Trees (0.926), and Histogram-based Gradient Boosting (0.934). The last section of Table 5 shows the AUROC scores of the different ML models trained on the given metadata.

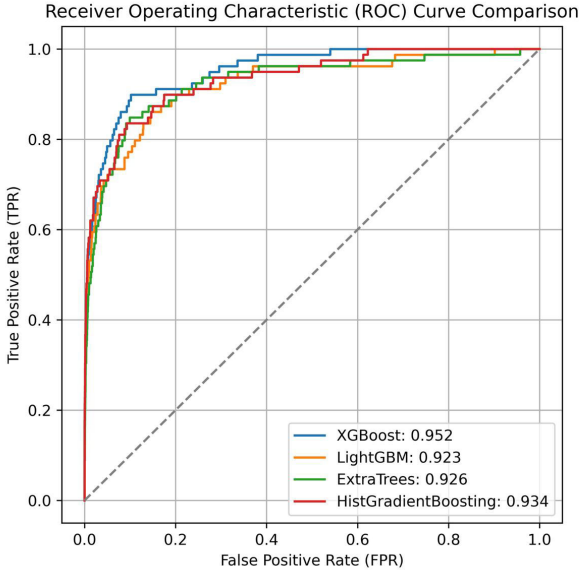


FIGURE 8. ROC curves for the various ML models.

XGBoost was able to outperform the other ML models on the clinical data due to several key factors. Its inherent ability to effectively handle tabular data, especially its ability to utilize derived features such as lesion color index, size, and position, demonstrates its efficacy in learning from such data. Moreover, its ability to robustly handle outliers, which are mostly part of the dataset due to potential errors during data collection, helps it learn from the data easily and effectively. Also, multifaceted diseases like Skin Cancer, often depend on non-linear relationships and inter-dependence of various factors, so tree-based models such as XGBoost are able to learn such relationships better than linear models. Further, the use of a weighted sampling strategy helped it overcome the huge class imbalance present in the dataset.

D. ENSEMBLE MODEL

The final ensemble model was formed using the weighted sum of the probabilities predicted by each of the models. SLSQP was used, which is often applied to get the optimal ensemble weights [45]. It was found that equal weights provided optimal performance on the validation set, returning an unweighted soft-voting ensemble classifier. Figure S1 provides the confusion matrices for each of the individual model as well as the ensemble model.

Fig. 9 compares the performance of the ensemble model with each of the three input streams. The difference in AUROC values of DeiT in Figures 4 and 9 is because of the difference in the test sets. The test set used in this figure is a subset of the dataset provided in the ISIC 2024 challenge, while the test set used in Fig. 4 is a combination of the test sets from all the datasets used in this study. The difference in AUROC scores for the two test sets can be attributed to the difference in the distribution of the test sets, the latter one being a larger superset of the prior. It can be noted that the ensemble model has a better balance of all the metrics, achieving an Accuracy of 91.76%, an AUROC of 0.953, a specificity of 91.77%, and a sensitivity of 84.81%. However, since the ISIC 2024 test set contains only 79 malignant cases, this sensitivity represents the correct classification of 67 cases, yielding a wide confidence interval (±7.26%). Although, DeiT is able to exhibit a superior performance in terms of sensitivity (96.10%), it has a much higher number of False Positives, resulting in a much reduced specificity on the combined test set. Further, it can be seen that the AUROC score for the ensemble model on the ISIC 2024 test set is better than all of its constituents, as evident from Fig. 9. The reason specificity becomes one of the crucial metrics, along with other metrics, is due to the huge imbalance present in the dataset, giving a much larger number of False Positives with a minute decrease in specificity. The optimal performance of the ensemble model can be attributed to the complementary nature of the individual models and the principle of ''wisdom of the crowd''.

To ensure that the image branch made any meaningful contribution to the ensemble model, an ablation study was conducted computing the ensemble using only the ANN and

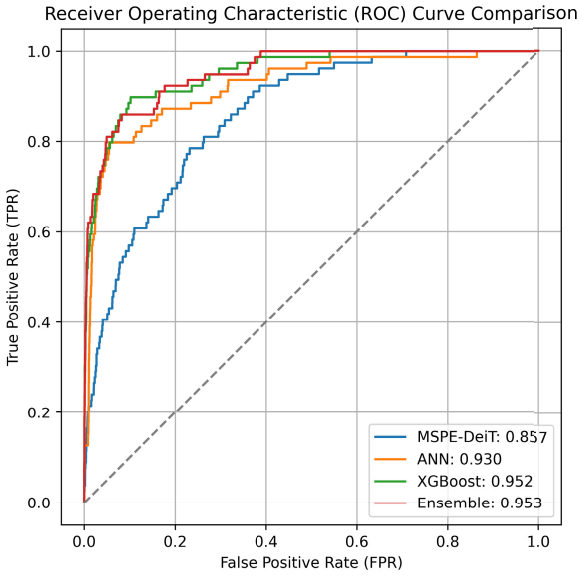


FIGURE 9. ROC curves for MSPE-DeiT, ANN, XGBoost, and Ensemble on the test set of ISIC 2024 dataset.

XGBoost and the AUROC for this was found to be same as the XGBoost, i.e., 0.952, showing the relative improvement made by the image branch on this subset. To further analyse the impact of the individual models on the performance of the ensemble model, we looked at the cases where the ensemble model disagreed with at least one of the base models. Table 6 shows the agreement across the three models. It can be noted that in the case of perfect agreement, the ensemble model is correct 99.88% of the time. It can be noticed that in about 0.12% of the cases, the ensemble provides an incorrect prediction, despite the individual models correctly classified those instances. This is because the soft-voting aggregation reduces the variance of the combined probability distribution, thus marginally pulling across the ensemble's independently optimized global threshold, where it sacrifices performance on some edge cases to provide globally superior perfor-

TABLE 6. Ensemble resolution capability across modality agreement cohorts.

Modality Agreement	N	% Dataset	Ensemble Correct	Ensemble Accuracy
All models correct	70,347	87.70	70,262	99.88%
MSPE-DeiT & ANN correct	5,228	6.52	3,036	58.07%
MSPE-DeiT & XGBoost correct	1,477	1.84	94	6.36%
XGBoost & ANN correct	191	0.24	183	95.81%
Only MSPE-DeiT correct	2,259	2.82	0	0.0%
Only XGBoost correct	23	0.03	4	17.39%
Only ANN correct	598	0.75	25	4.18%
All models incorrect	89	0.11	0	0.0%
Total	80,212	100.0	73,604	91.76%

TABLE 7. Comparison of various models on skin lesion classification datasets.

Ref.	Image data	Clinical data	Multimodal	Model	Accuracy (%)	AUROC	Specificity (%)	Sensitivity (%)
[13]	ISIC 2018	✗	✗	CNN, Resnet50, InceptionV3, Inception Resnet	CNN: 83.20, Resnet50: 83.70, InceptionV3: 85.80, Inception Resnet: 84.00	–	–	–
[19]	ISIC 2019	✗	✗	Modified GoogleNet	94.92	–	97.00	79.80
[14]	ISIC 2018, ISIC 2019, ISIC 2020	✗	✗	InSiNet	ISIC 2018: 94.59, ISIC 2019: 91.89, ISIC 2020: 90.54	–	ISIC 2018: 91.18, ISIC 2019: 94.12, ISIC 2020: 88.64	ISIC 2018: 97.50, ISIC 2019: 90.00, ISIC 2020: 93.33
[15]	ISIC 2020, HAM10000, DermIS	✗	✗	Deep learning-based skin cancer classification network (DSCC_Net)	94.17	0.994	94.28	93.76
[22]	ISIC 2018, ISIC 2020, Kaggle Dataset	✗	✗	Multi-head attention block depthwise separable convolution network	78.21	0.785	72.36	84.68
[23]	HAM10000, Private dataset	✗	✗	SkinTrans	HAM10000: 94.30	HAM10000: 0.987	–	–
[24]	HAM10000	✗	✗	ViTSCD-Large	94.10	–	–	–
[25]	ISIC 2019	✗	✗	Hybrid Swin Transformer and SwiGLU-based MLP	89.36	–	–	85.13
[16]	HAM10000	✗	✗	GRU + Improved Orca Predation Algorithm	99.00	–	97.00	95.00
Proposed	ISIC 2016, ISIC 2017, ISIC 2018, ISIC 2019, ISIC 2020, ISIC 2024	✗	✓	MSPE-DeiT	89.05 ± 0.20	0.965 ± 0.003	88.89 ± 0.20	96.10 ± 0.98
	✗	ISIC 2024	✗	ANN	95.20 ± 0.15	0.930 ± 0.038	95.22 ± 0.15	78.48 ± 8.91
	✗	ISIC 2024	✗	XGBoost	89.81 ± 0.23	0.952 ± 0.022	89.81 ± 0.23	89.87 ± 6.78
	ISIC 2016, ISIC 2017, ISIC 2018, ISIC 2019, ISIC 2020, ISIC 2024	ISIC 2024[1]	✓	Ensemble	91.76 ± 0.19	0.953 ± 0.020	91.77 ± 0.20	84.81 ± 7.26

[1] Only ISIC 2024 data was used for testing the ensemble model due to the unavailability of clinical data in the other datasets. Performance metrics are reported along with 95% confidence intervals obtained using bootstrapping with 1000 iterations.

mance [48]. Further, the ensemble model correctly classified 3,036 instances where-in MSPE-DeiT and the ANN models were correct and had a disagreement with XGBoost. This shows the contribution of these two models towards the ensemble decision playing a key role in improving the performance of the ensemble model. Similarly, in cases where MSPE-DeiT was incorrect, while ANN and XGBoost were correct, the ensemble predicted correctly 95.81% of the time. Furthermore, the lower accuracies in case of single correct model (0% in case of MSPE-DeiT) show that the ensemble doesn't alter its decision based on a single model's prediction and is robust to such cases. This demonstrates the efficacy of the ensemble model, benefitting from all the three individual models' complementary strengths.

MSPE-DeiT processes image data, leveraging self-attention mechanisms to capture intricate visual patterns. In contrast, ANNs and XGBoost operate on clinical metadata, with XGBoost employing a tree-based approach and ANNs utilizing complex layered structures to learn the underlying relationships. This diversity in model architectures and input modalities ensures that they capture different aspects of the data and learn distinct patterns. By employing a soft-voting classifier, the ensemble benefits from the collective intelligence of these uncorrelated models. Even if one model misclassifies a particular instance, the others can often compensate, leading to a more robust prediction. Furthermore, while individual models might be susceptible to overfitting specific patterns, the combined ensemble helps mitigate this risk by computing a weighted average. This harmonious combination of diverse models, each with its unique strengths, results in a more accurate and generalizable prediction.

E. SoTA COMPARISON

Table 7 compares all the proposed methods to the existing state-of-the-art results. The results presented for the ensemble model in the table correspond to the ISIC 2024 test set specifically, since the metadata associated with the other datasets wasn't available, hence the approach can't be tested across the other datasets. All the performance evaluation metrics for the proposed models have been computed along with the 95% confidence interval using bootstrapping with 1000 iterations.

Most of the past studies [13], [16], [19], [24], [25] have used only a single dataset, highlighting the inherent challenge of a smaller dataset size and an increased chance of overfitting, limiting generalization on other populations. Though researchers like [14], [15], [22], and [23] worked on diverse datasets, all of them relied only on dermoscopic images. In contrast, the proposed approach not only incorporates the dermoscopic images but also the 3D-TBP scans and the associated metadata, making it a much more robust approach. The use of such a multimodal and diverse dataset incorporating various publicly accessible datasets increases the generalizability of the approach as well as its robustness to errors. Also, the inherent capability of ViTs in capturing global contextual information compared to CNNs used in most of the studies show the added advantage of the study. The modality-specific embeddings, followed by training of a shared encoder, helped MSPE-DeiT to learn cross-modal relationships, thus giving it a large data corpus to train. Moreover, the image information is accompanied by the lesion characteristics in the clinical data, increasing the model's access to information and aiding in decision-making.

F. COMPUTATIONAL COST ANALYSIS

To evaluate the model's efficiency, we assessed its training time, inference time, memory footprint, and parameter count. Table 8 displays the computational costs of MSPE-DeiT, XGBoost, ANN, and the Ensemble model. All these metrics were computed on a Kaggle T4 GPU to ensure consistency and convenient reproducibility of all the results. Further, the inference time was computed for 500 batches for each model and the average inference time has been reported, using an initial warmup runs for 20 batches each to avoid any inflation due to those. The memory footprint reported is based on the peak memory usage for the entire duration. It can be observed that XGBoost has the least training time, inference time, and parameter count. However, its memory footprint is much larger than the ANN model, while having much lesser number of parameters than the ANN model. This can be attributed to the fact that XGBoost is a tree-based model and requires loading its complete tree structure for inference, which leads to a larger memory footprint compared to the lightweight ANN model, utilizing only the weights and the biases. On the other hand, MSPE-DeiT took the most training time, inference time, memory footprint, and the number of parameters. This can be attributed to the inherent complexity associated with transformer-based architectures and the need for training multiple components, i.e., multi-head self-attention blocks, feed-forward networks, and modality-specific embedding layers, which require substantial computational resources. The ensemble model, being a combination of all the three models, took the most training time, memory footprint, and number of parameters. Despite these limitations, the ensemble is able to provide the inference for an input in just 6.2 ms, processing more than 160 inputs in a second, proving it to be fit for deployments in real-world clinical scenarios.

TABLE 8. Computational cost analysis for the various models used in the study.

Model	Training time (min)	Inference time (ms)	Memory footprint (MB)	#params
MSPE-DeiT	390.72	5.370	64.10	56,72,450
XGBoost	0.15	0.005	19.28	16,169
ANN	0.91	0.795	0.21	52,058
Ensemble	391.78	6.200	83.59	5,740,677

Further, the peak memory footprint is only 83.59 MB, which is well within the typical memory constraints of most modern edge devices.

V. CONCLUSION

The goal of this study was to improve the diagnostic accuracy of skin cancer categorization by integrating image-based and tabular clinical data into a multimodal deep-learning architecture. ANNs for clinical feature representation, MSPE-DeiT for unpaired modality lesion image analysis, and XGBoost for structured clinical data categorization were the three different modeling approaches that were examined. The clinical metadata were used as provided in the ISIC 2024 dataset, and image data was gathered from several publicly accessible datasets, namely ISIC 2016, ISIC 2017, ISIC 2018, ISIC 2019, ISIC 2020, and ISIC 2024, to enhance predictive performance. Each model performed quite well on its own in the classification assignment.

An ensemble strategy that combined the feature extraction powers of MSPE-DeiT, the representational effectiveness of ANNs, and the structured data handling efficacy of XGBoost was used to capitalize on the complementary characteristics of these models. By fine-tuning the ANN's hyperparameters using Bayesian optimization, vanishing gradient problems were successfully avoided, and convergence was enhanced. The findings highlight the multimodal integration's potential for practical clinical applications by showing that it greatly improves diagnostic robustness and accuracy.

Even though the suggested framework shows encouraging outcomes, there are still several areas that require further study. A key limitation of the current study is that the multimodal ensemble evaluation was restricted to the ISIC 2024 dataset, as clinical metadata was unavailable for the older datasets (ISIC 2016-2020). Consequently, while the framework demonstrates strong performance on the 2024 data, its multimodal generalizability across the older cohorts remains to be fully established. To reduce dependency on human-supplied clinical variables, one important topic is the development of automated feature extraction approaches to extract metadata directly from dermoscopic and 3D-TBP images. To provide greater accessibility, efforts should also be focused on the real-time implementation of this paradigm in clinical practice, especially through telemedicine applications and smartphone-based screening tools. To evaluate the suggested approach's generalizability and dependability across a range of demographics and skin types, more validation through extensive clinical trials is necessary.

Additionally, incorporating explainability techniques like SHAP values for tabular data or Grad-CAM for MSPE-DeiT should improve the interpretability of model predictions, leading to increased acceptance and trust in clinical decision-making. Diagnostic performance could be further enhanced by incorporating additional modalities, such as histopathological and genomic data, and broadening the scope of this framework to classify a wider range of dermatological disorders. These developments would strengthen the importance of machine learning in precision medicine by supporting the continued development of AI-driven dermatological diagnostics.

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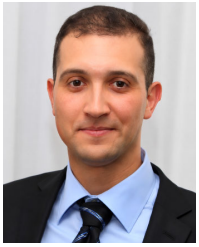
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